

NATIONAL UNIVERSITY OF SINGAPORE

MA1513 – EXAM (90 minutes)

AY 2022/2023 Semester II

INSTRUCTIONS TO STUDENTS

- (a) There are 15 MCQ and 4 written questions in this exam with a total of 90 marks.
- (b) The MCQ should be answered and submitted through Exemplify.
- (c) For the written questions, use the exam answer booklet to write the answers. (Writing on separate electronic device is not allowed).
 - Write down your student number clearly on the cover page of the answer booklet.
 - Write the question number (LQ1 to LQ4) on every written page of the answer booklet.
 - Scan or take pictures of all the written pages (make sure the images can be read clearly). Merge all your images into one pdf file (arrange them in order of the questions (LQ1, LQ2, LQ3, LQ4). Name the pdf file by your student number and module code (e.g. A123456R_MA1513.pdf).
 - At the end of the exam, you have 15 minutes to upload your pdf into the CANVAS folder “Exam submission”.
 - Your answer booklet will be collected for reference. Grading will be based on the PDF version.
- (d) Students are not allowed to exit Exemplify before the end of the exam. At the end of the 90 minutes, Exemplify will close automatically, and you should immediately stop writing and prepare to upload your written answers.
- (e) This is an open book examination.
- (f) You may use MATLAB and any calculator. However, you should explain clearly how the answers of the written questions are obtained.

MA1513 Exam (Exemplify)

AY2022/2023 Semester II

MCQ/FIB (15 questions; 3 marks each)

1. Given the row echelon form $\left(\begin{array}{ccc|c} a & 1 & 1 & 1 \\ 0 & b & 1 & 1 \\ 0 & 0 & c & 1 \end{array}\right)$ of an augmented matrix for a linear system with a, b, c being some real numbers. Which of the following statements are true? (Pick all correct options)
 - a. This system is consistent for any values of a, b, c .
 - b. If the system is consistent, the system has exactly one solution.
 - c. It is possible that the system has infinitely many solutions.
 - d. The first column of the augmented matrix can be a non-pivot column.
 - e. The augmented matrix is full rank for any values of a, b, c .

2. Given that A and B are non-singular $n \times n$ matrices, which of the following statements are always true? (Pick all correct options)
 - a. AB is non-singular.
 - b. $A + B$ is non-singular.
 - c. $A^{-1}B^{-1} = (AB)^{-1}$
 - d. The transpose A^T of A is non-singular.
 - e. cB is non-singular for any scalar c .

3. Which of the following determinant has the same value as $\begin{vmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{vmatrix}$ for any values of a, b , and c ? (Pick all correct options)
 - a. $\begin{vmatrix} a & 0 & 0 \\ a & b & 0 \\ a & b & c \end{vmatrix}$
 - b. $\begin{vmatrix} 0 & 0 & c \\ 0 & b & c \\ a & b & c \end{vmatrix}$
 - c. $\begin{vmatrix} 0 & c & 0 \\ b & c & 0 \\ b & c & a \end{vmatrix}$
 - d. $\begin{vmatrix} b & a & c \\ 0 & 0 & c \\ b & 0 & c \end{vmatrix}$
 - e. $\begin{vmatrix} a & 0 & 0 \\ b & b & 0 \\ c & c & c \end{vmatrix}$

4. Which of the following linear spans contain the vector $(3,6,9)$? (Pick all correct answers.)
 - a. $\text{Span}\{(9,0,0), (0,6,0), (0,0,3)\}$
 - b. $\text{Span}\{(9,6,3)\}$
 - c. $\text{Span}\{(10,20,30)\}$
 - d. $\text{Span}\{(3,0,9), (0,6,0)\}$
 - e. $\text{Span}\{(1,1,0), (0,1,1), (1,2,1), (1,0,-1)\}$

5. Which of the following sets of vectors are bases for $\mathbf{R}^4$? (Pick all correct answers)
- $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (1, 1, 1, 1)$
 - $(1, 1, 1, 1), (1, 0, 1, 0), (0, 1, 0, 1), (0, 1, 1, 0)$
 - $(1, 2, 1, 1), (1, 1, 2, 1), (1, 1, 1, 2)$
 - $(1, 2, 3, 1), (1, 4, 5, 1), (1, 6, 7, 1), (1, 8, 9, 1)$
 - $(1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), (0, 1, 1, 1)$
6. The projection of the vector $(3, 3, 3)$ onto the plane with equation $x - y + 2z = 0$ is given by $p = (a, b, c)$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)
7. Let M be a 3×3 non-singular matrix with columns $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ in that order. Suppose $M \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$. Which of the following statements are true? (Pick all correct answers)
- $3\mathbf{c}_1 + 2\mathbf{c}_2 + \mathbf{c}_3$ is the coordinate vector of $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ with respect to the basis $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{R}^3$.
 - $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ is the coordinate vector of $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ with respect to the basis $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{R}^3$.
 - $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ is the coordinate vector of $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ with respect to the basis $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{R}^3$.
 - $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is the coordinate vector of $\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{c}_3$ with respect to the basis $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{R}^3$.
 - $\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{c}_3$ is the coordinate vector of $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ with respect to the basis $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ of $\mathbf{R}^3$.
8. Suppose an unknown A is singular and its reduced row echelon form R is known. Which of the following statements are true? (Pick all correct answers)
- We can determine the row space of A .
 - We can determine the column space of A .
 - We can determine the nullspace of A .
 - We can determine the matrix A .
 - We can find a basis for the solution space of $Ax = 0$.
9. (Linear transformation) Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation with standard matrix given by $A = \begin{pmatrix} 1 & 2 & a \\ 2 & 0 & b \\ 1 & -1 & c \end{pmatrix}$. Suppose $T \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$. Then $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

10. In the following 3x3 matrices, each * can represent any number.

Which of the following matrices must be diagonalizable? (Pick all correct answers)

a. $A = \begin{pmatrix} 1 & 0 & 0 \\ * & 3 & 0 \\ * & * & 5 \end{pmatrix}$

b. $B = \begin{pmatrix} 6 & * & * \\ 0 & 0 & * \\ 0 & 0 & 6 \end{pmatrix}$

c. $C = \begin{pmatrix} * & 1 & 2 \\ 1 & * & 3 \\ 2 & 3 & * \end{pmatrix}$

d. Matrix D has only 1 eigenvalue and the eigenspace is given by $\mathbf{R}^3$

e. Matrix E can be reduced by Gaussian elimination to a diagonal matrix.

11. Given

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \quad \text{for some real numbers } a \text{ and } b. \text{ Which of the}$$

following statements are true? (Pick all correct answers)

a. $A = \begin{pmatrix} 7 & 0 & -1 \\ 7 & 0 & 0 \\ 7 & 3 & -1 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} 7 & 0 & -1 \\ 7 & 0 & 0 \\ 7 & 3 & -1 \end{pmatrix}^{-1}$

b. $2A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 0 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2a & 0 & 0 \\ 0 & 2a & 0 \\ 0 & 0 & 2b \end{pmatrix} \begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 0 \\ 2 & 2 & 2 \end{pmatrix}^{-1}$

c. $A^3 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^3 \begin{pmatrix} a^3 & 0 & 0 \\ 0 & a^3 & 0 \\ 0 & 0 & b^3 \end{pmatrix} \left(\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \right)^3$

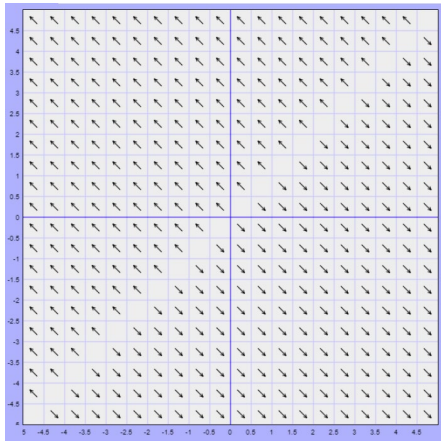
d. $A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1/a & 0 & 0 \\ 0 & 1/a & 0 \\ 0 & 0 & 1/b \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

e. $A = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} b & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}^{-1}$

12. The general solution of an SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ is given by

$$y = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}. \text{ Then } a = _, b = _, c = _, d = _ \text{ (FIB)}$$

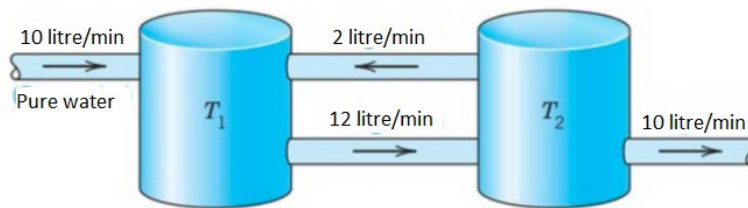
13. The direction field of a 2x2 SDE $y' = Ay$ is given below.



Which of the following statements are true? (Pick all correct answers)

- a. A is singular
- b. A is not diagonalizable
- c. $\text{span}\{(1,1)\}$ is an eigenspace of A
- d. The only equilibrium point of the SDE is at the origin
- e. A has a positive eigenvalue

14. Consider the two tanks shown in the diagram below. Each tank initially holds 100 litre of salt water. The amount of salt at time t (min) in the two tanks are $y_1(t)$ and $y_2(t)$ (kg) respectively.



The SDE that solves for y_1 and y_2 is given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$ (FIB) (Give your answers in decimal form with 2 decimal place)

15. Given the SDE $\begin{cases} y_1' = 2y_1 + y_2 \\ y_2' = -y_1 + 2y_2 \end{cases}$

Which of the following statements are true? (Pick all correct answers)

- a. A real solution of the SDE is $e^{2t} \begin{pmatrix} \cos t + \sin t \\ \cos t - \sin t \end{pmatrix}$.
- b. $e^{2t} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$ is not a real solution of the SDE.
- c. The matrix A has an eigenvalue $2 + i$ with corresponding eigenvector $\begin{pmatrix} 1 \\ -i \end{pmatrix}$.
- d. The equilibrium point of the SDE is unstable.
- e. The trajectories of the SDE are counter-clockwise.

Long Questions (4 questions; total 45 marks)

Long Question 1 (LQ1)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ1 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

An electrical appliance company produces 3 models (A, B, C) of vacuum cleaners. The amount (in terms of dollar \$) of required resources to produce each unit of vacuum cleaners are given below:

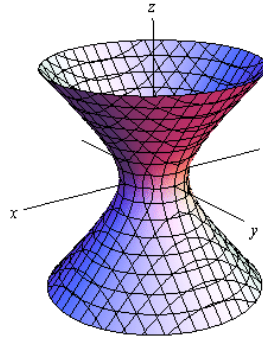
	Labour	Material	Miscellaneous
A	30	10	20
B	30	20	30
C	40	20	20

- (i) [3 marks] Suppose the company wants to produce 300 units of A, 400 units of B and 500 units of C. Express the budget needed for labour, material and miscellaneous items as a product of a 3×3 matrix with a 3×1 column vector, and write down the three amounts.
- (ii) [3 marks] Suppose the budget of the company are: \$40,000 on labour, \$20,000 on material and \$30,000 on other miscellaneous items. Determine the number of units of each model of vacuum cleaners that should be produced if the company wants to use up all of its budget in its production schedule.
- (iii) [4 marks] Suppose the company decides to produce one more model D of vacuum cleaners with the same budget as (ii). The labour, material and miscellaneous cost for producing one unit of D are \$40, \$30 and \$30 respectively. Find the largest number of units of D that can be produced if the company wants to use up all of its budget in its production schedule.

Long Question 2 (LQ2)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ2 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

The following surface in the 3D space is known as a hyperboloid. The horizontal cross sections are circles of different radii.



To determine the equation of the surface, which has the form

$$ax^2 + ay^2 + bz^2 = 1 \quad (*)$$

the coordinates of several points at various height on the surface were measured: (2.65, 1.53, 1), (2.80, 1.62, 2), (3.03, 1.75, 3), (3.33, 1.92, 4), (3.67, 2.12, 5).

- (i) [3 marks] Set up a 5 x 2 linear system $Ax = B$ in variables a and b , by substituting the given data points into equation (*).
- (ii) [4 marks] Find the least squares solution of $Ax = B$ and write down the best fit equation for the surface. Give your answer up to 4 decimal places.
- (iii) [3 marks] Use the equation in (ii) to estimate the cross sectional radius of the hyperboloid at $z = 10$. Give your answer up to 4 decimal places.

Long Question 3 (LQ3)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ3 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

Three telecommunication companies X, Y, Z offering similar mobile plans with some unique features in each plan. According to market research, for each month, there are movements among the clients of the three companies: 2% and 4% of clients of X switch to Y and Z respective; 3% and 3% of clients of Y switch to X and Z respective; 4% and 2% of clients of Z switch to X and Y respective.

- (i) [3 marks] Let $v_n = \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix}$ denote the market share of the three companies in the n th month. Find the matrix A such that $v_{n+1} = Av_n$.
- (ii) [4 marks] Write down the n -th power A^n of A in the form PDP^{-1} where D is a diagonal matrix.
- (iii) [3 marks] What is the market share of the three companies in the long run? Does it stabilise?
- (iv) [2 marks] How is the long term market share related to the stage matrix A ?

Long Question 4 (LQ4)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ4 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

A damped mass-spring system is given by the second order ODE

$$my'' + cy' + ky = 0 \quad (*)$$

where m is the mass (in kg) of the ball, k is the spring constant (in Newton/m) and c is the damping constant. Suppose the ball is 4 kg and the spring constant is 4 Newton/m.

- (i) [2 marks] Suppose the ball returns to equilibrium immediately without oscillation. Find the damping constant c .
- (ii) [5 marks] Convert the ODE (*) with the c in (i) to a 2 x 2 SDE and find a fundamental set of solutions of the SDE.
- (iii) [3 marks] Sketch the phase portrait of the SDE in (ii) indicating the directions of the trajectories.
- (iv) [3 marks] Explain why the equilibrium point of the SDE that describe any damped mass-spring system cannot be a saddle point regardless of the value of the damping constant c .